

Dian Jin

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EDUCATION

John Hopkins University

M.S. in Computer Science.

University of Wisconsin - Madison

B.S. in Computer Science; B.S. in Data Science, 2020 Dean's List.

Baltimore, MD

Aug 2025 - Present

Madison, WI

Aug 2021 - May 2024

PROGRAMMING SKILLS

Skills: Python, C, C++, JavaScript, CSS, C#, Java, HTML, Django, Flask, AWS Cloud, MQTT, Vue.js, React.js, Node.js, langchain, LLM, RAG, MySQL, PostgreSQL, MongoDB, SQLite3, UI/UX, Jenkins, Git, Android and iOS, Unity

Field: Embedded System, Operating System, Big Data, Data Structure, Algorithm, Machine Learning, Neural Network, frontend and backend development, API integrations

Others: Game-play Development, Project Proposals, work with scrum teams,

WORK EXPERIENCE

John Deere

Software Product Engineer

May 2024 - Present

Rotation 2

- **Led full-stack development of a Django web application** that captures real-time data from embedded C++ controllers, transmits it to AWS SQL, and visualizes it via Kinesis streams—reducing latency by 40% and enabling timestamp-based cost analysis.
- **Designed and implemented a cost-efficiency model** that helped leadership optimize data processing windows, resulting in 25% cloud budget savings and improved decision-making on real-time data usage.
- **Built and globally deployed a diagnostic console** using Django and Kinesis to monitor embedded controller errors. Integrated C++ modules decode CAN streams and transmit diagnostics to the cloud, increasing engineering efficiency by 60% and eliminating on-site machine checks.
- **Architected a customer-facing web platform** to visualize machine paths using silo compaction maps. Extracted and decoded embedded CSV data, transformed it into SQLite3, and rendered NumPy grids as TIFF images—earning 90%+ positive feedback from users.
- **Served as technical lead and strategic architect** across all projects, driving system design, implementation, and cross-functional collaboration with cloud, embedded, and data teams.

Rotation 1

- **Developed a Status Detection module** for display software to monitor and optimize machine performance in real time.
- **Built full stack components** using C++ and Qt, interfacing with embedded systems to capture live signals via CAN (Controller Area Network).
- **Integrated the module into production pipelines**, deploying it as a software artifact for real-time visualization and user interaction.
- **Maintained and extended system infrastructure**, including the signal table, SQL database, CSV logging, and Shell scripts, ensuring reliable data handling and traceability across releases.
- **Enabled real-time automation and diagnostics** by decoding controller signals and integrating them into backend processing and display overlays.
- **Designed and implemented a designer tool** to analyze software deliveries on a quarterly basis and integrate the tool for generating for all production needs, Integrated API and data parsing, deployed globally to software teams using Python. Achieved a 30% increase in positive customer feedback and a 20% rise in usage, for products. Delivering enhancements for Product.

John Deere

Front-end Software Engineer Intern

May 2023 - Aug 2023

- **Led front-end development** of an automated path planning application using C++ and Qt, enabling users to shift machine paths interactively through the UI.
- **Implemented coordinate conversion algorithms** to translate user-defined meter shifts into global latitude and longitude, ensuring precise geolocation across environments.

- **Integrated real-time backend communication with SQL databases**, allowing seamless unit switching and dynamic updates based on user input.
- **Designed intuitive UI components** for defining complex workflows and schedules, improving usability and reducing setup time.
- **Collaborated with automation and algorithm teams** to calibrate path adjustments and ensure accurate execution across embedded systems.
- **Achieved a 97% feature certification rate** and received strong internal feedback for improving customer flexibility and operational control.

Ark Technology

IOS Developer Intern

Nov 2021 - Jan 2022

- **Designed and developed a modern iOS social platform** using Vue.js, React.js, HTML, and CSS, delivering a responsive and engaging user interface.
- **Translated Figma-based UX designs** into functional front-end components, improving interface quality and user interaction flow by 35%.
- **Conducted thorough testing and QA across multiple modules**, proactively resolving technical issues before beta release.
- **Launched the app successfully on the App Store**, attracting nearly 1,000 users on day one and achieving a 40% increase in engagement within the first week.
- **Enhanced team productivity** by streamlining workflow definitions and automating task scheduling through intuitive UI design.

PROJECT

Software Lead

Minerva Game Studio

Sep 2022 - Present

Madison, WI

- Developed a 2D video game for PC platforms using **Unity, C#** with friends, Published in steam.
- Seamlessly integrated in-game **AI**, developed the movement scripts for both NPCs and Boss encounters, enhancing the overall game play experience.
- Successfully rolled out the beta version within the academic environment, garnering acclaim from 90% of players.
- Demonstrated exceptional coding expertise by optimizing the code base, reducing time complexity by more than 50%.
- Demo: https://store.steampowered.com/app/3102950/Library_of_Meialia.

OS development & AI development

Feb 2022 - May 2022

- Developed a custom **Linux kernel** for Intel IA-32 architecture using **C** and **x86 assembly**.
- Implemented filesystem with Paging algorithm to successfully enable the execution of various files and storage.
- Develop a program capable of recognizing digits in images using Convolutional Neural Networks (**CNN**) and **Neural Network** techniques in **Python**.

HONORS & AWARDS

Dean's list

Academic Excellence Scholarship of Honors School

Heart of Illinois Scholarship

UW-Madison, Fall 2021

School of Honor, Every Semester

Heart of Illinois, Every Semester